

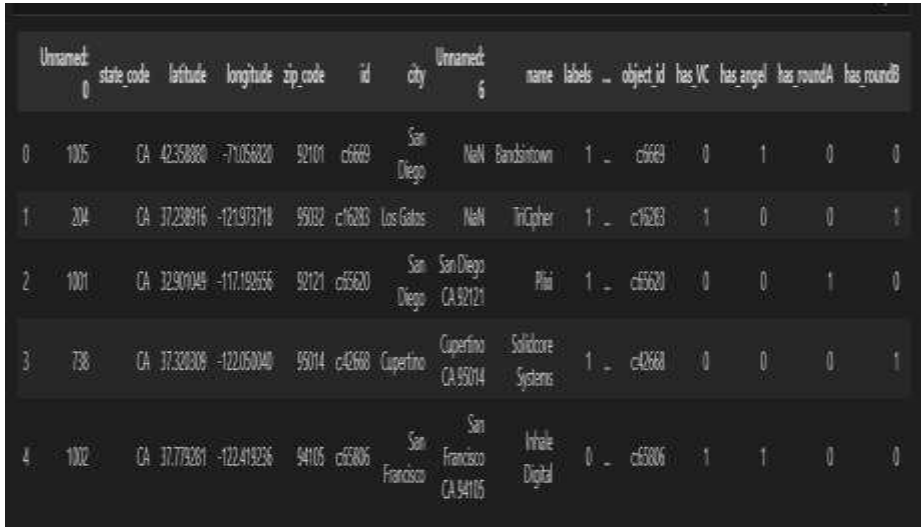
Data Collection and Preprocessing Phase

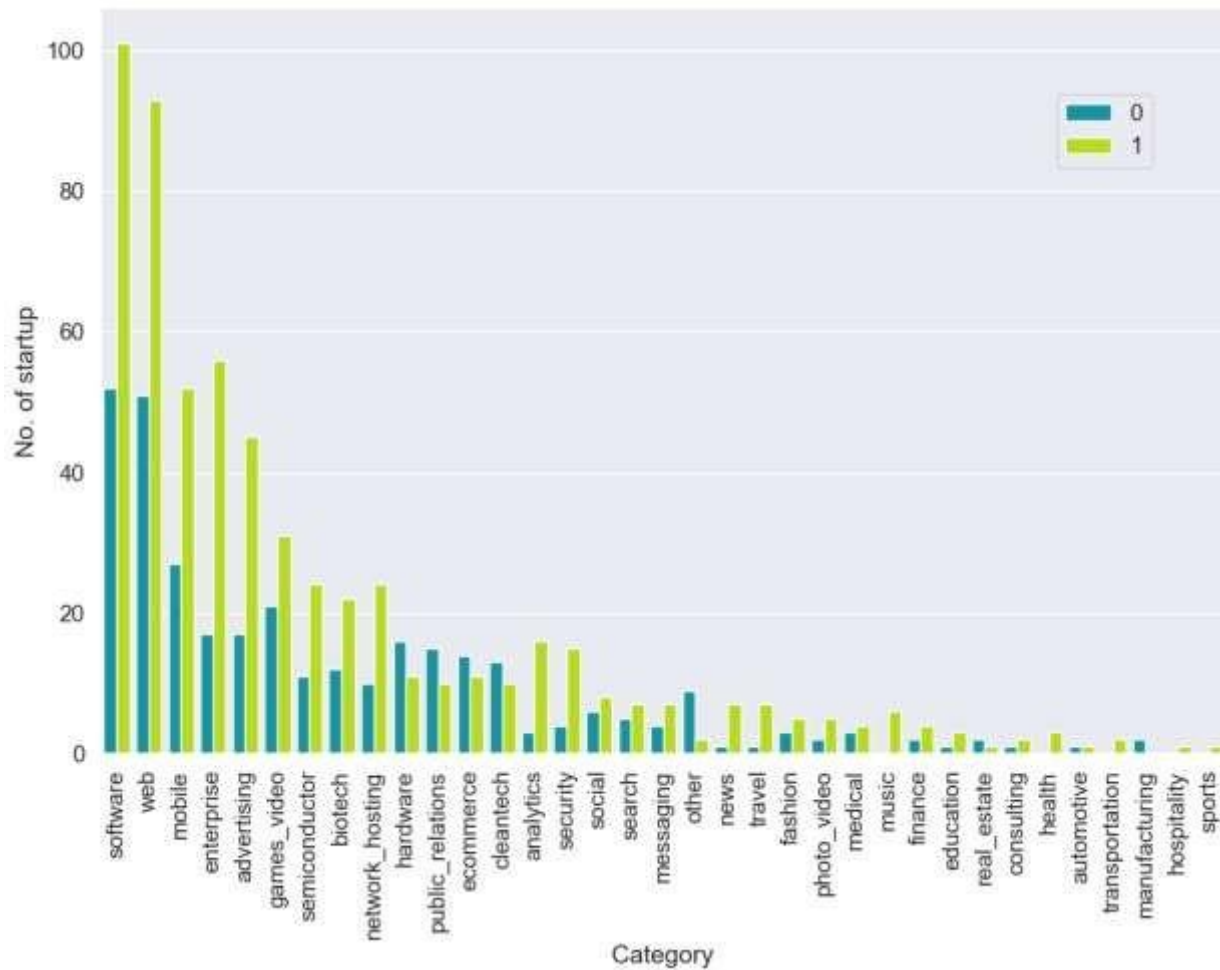
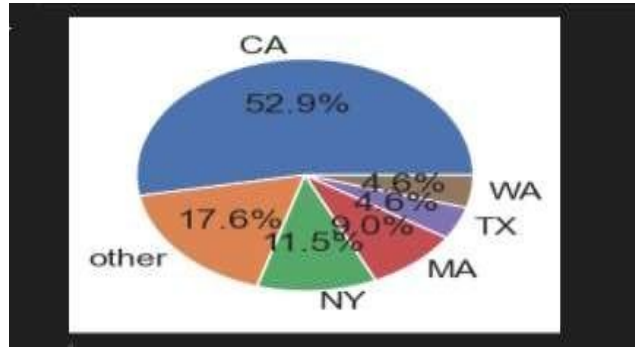
Date	20-02-2026
Team ID	LTVIP2026TMIDS62277
Project Title	prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	6 Marks

Data Exploration and Preprocessing Report

Dataset variables will be statistically analyzed to identify patterns and outliers, with Python employed for preprocessing tasks like normalization and feature engineering. Data cleaning will address missing values and outliers, ensuring quality for subsequent analysis and modeling, and forming a strong foundation for insights and predictions.

Section	Description
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	<u>Dimension:</u> 923 rows × 49 columns <u>Descriptive statistics:</u>
Data Overview	
Univariate Analysis	



Bivariate
Analysis

Outliers and
Anomalies

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Data Preprocessing Code Screenshots

Loading Data

```
data = pd.read_csv('startup.csv')
data
```

	Unnamed: 0	state_code	latitude	longitude	zip_code	id	city	Unnamed: 6	name	labels	...	object_id	has_VC	has_angel	has_roundA	has_roundB
0	1005	CA	42.356601	-71.056620	02101	c5668	San Diego	NaN	Sandtown	1	-	c5668	0	1	0	1
1	294	CA	32.28994	-121.972758	95032	c5623	Los Gatos	NaN	HiGphre	1	-	c5623	1	0	0	1
2	1001	CA	32.901049	-117.39336	92121	c5620	San Diego	San Diego CA 92121	Plai	1	-	c5620	0	0	1	1
3	738	CA	37.320389	-122.030940	95014	c42668	Capetino	Capetino CA 95014	Soldcore Systems	1	-	c42668	0	0	0	1
4	1002	CA	37.779281	-122.419236	94105	c85906	San Francisco	San Francisco CA 94105	Yrhole Digital	0	-	c85906	1	1	0	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
918	352	CA	37.46284	-122.376471	94102	c21343	San Francisco	NaN	Colacort	1	-	c21343	0	0	1	1
919	721	MA	42.594817	-71.295811	1003	c41747	Durington	Burlington MA 1003	Red Point Systems	0	-	c41747	1	0	0	1
920	557	CA	37.483281	-122.075303	94089	c31549	Sunnyvale	NaN	Paraco Medical	0	-	c31549	0	0	0	1
921	584	CA	37.56732	-122.260726	94034	c33198	San Francisco	NaN	Carasta	1	-	c33198	0	0	1	1
922	482	CA	37.386778	-121.942777	93554	c25702	Santa Clara	Santa Clara CA 95054	Aurepra Technologies	1	-	c25702	0	0	0	1

Handling Missing Data

```
#filling missing value column(unnamed:6)
data['Unnamed: 6'] = data.apply(lambda row: (row.city) + " " + (row.state_code) + " " + (row.zip_code) , axis = 1)

# Total Missing Values column "Unnamed: 6"
totalNull = data['Unnamed: 6'].isnull().sum()

print('Total Missing Values kolom "Unnamed: 6": ', totalNull)

#filling missing values of column(closed_at)
data['closed_at'] = data['closed_at'].fillna(value="31/12/2013")
totalNull = data['closed_at'].isnull().sum()

print('Total Missing Values kolom "closed_at": ', totalNull)
```

Data Transformation	<pre>data["status"] = data.status.map({'acquired':1, 'closed':0}) data["status"].astype(int)</pre> <pre>0 1 1 1 2 1 3 1 4 0 ... 918 1 919 0 920 0 921 1 922 1 Name: status, Length: 923, dtype: int64</pre>
Feature Engineering	Attached the codes in final submission.
Save Processed Data	-